

SnapShield

Private multimodal security for the AI PC era.

See it. Analyze it. Protect it — privately.

The problem

Security attacks increasingly combine messages, links, fake interfaces, documents and QR codes. Cloud AI can add latency and exposes sensitive security content to external services.

The solution

SnapShield is a local-first security intelligence layer for Snapdragon-powered Windows PCs. It analyzes suspicious content with a real pretrained phishing classifier, URL/security evidence and an explainable risk-fusion layer.

Why Snapdragon

The product is designed around heterogeneous compute: CPU orchestration, GPU/visual workloads where useful, and the Hexagon NPU for supported AI inference. Qualcomm AI Hub provides optimized model pathways and Snapdragon compute targets.

Technical implementation

Current working path: input → local transformer → security evidence → risk fusion → recommendation. The inference layer is modular for ONNX/QNN deployment. Runtime detection prevents the application from falsely claiming NPU execution.

Multimodal roadmap

Screenshot/PDF/QR inputs share the same security-evidence layer. Local OCR and vision adapters can be attached without changing the decision architecture.

Privacy

Text analysis requires no cloud AI API. The design principle is simple: sensitive security content should stay on the device whenever the selected inference path is local.

Evaluation plan

Technical: model quality, latency, memory and NPU execution. Innovation: multimodal security designed for the AI PC. Deployment: local Windows application with modular Snapdragon runtime. Presentation: concise evidence-led demo and reproducible repository.

Winning thesis

SnapShield treats the AI PC itself as part of the security architecture—not merely as a faster place to run a cloud client.